



Anusara Karunarathna

✉ anusarakarunaratna@gmail.com ☎ +9471-544-2990 📍 Colombo

🌐 <https://www.linkedin.com/in/anusara-karunarathna-359443222/>

🐙 <https://github.com/AnusaraKarunaratna> 🔗 anusara.dev

📄 PERSONAL STATEMENT

Strong Software Engineer proficient in full-stack development, API design, database optimization with experience in cloud-native engineering, backend performance tuning, and enterprise-grade software development practices.

💼 EXPERIENCE

04/2024 – 09/2025
Colombo

SLT Mobitel - Intern & Freelance Software Engineer

- Developed and enhanced the Sri Lanka Telecom Business Continuity Management System as an enterprise backend platform using Node.js and Express.js, implementing MVC architecture with scalable service orchestration and complex API call-tree communication.
- Implemented secure JWT-based authentication and authorization, including token lifecycle management, middleware validation, and role-based access control.
- Designed and optimized RESTful APIs and internal database operations, handling complex queries, relational data processing, and performance-tuned access patterns for enterprise workloads.
- Engineered enterprise integrations for Alliance Finance and INSEE Cement Lanka, incorporating OCR-based data extraction, API-driven processing, and system-to-system automation in Agile environments with version control via Bitbucket.

🎓 EDUCATION

01/2023 – 05/2026

BEng (Hons) Software Engineering,
Informatics Institute of Technology affiliated with the University of Westminster
First Class Honours (80%+ average)

06/2022 – 12/2022

Foundation Certificate in Higher Education, Informatics Institute of Technology
Grade Distinction

01/2020 – 05/2022

GCE O-Level, Ananda College Colombo
9*As

🧠 SKILLS

Programming & Scripting — Java, Python, C#, JavaScript, PHP

Backend Engineering — Node.js, Express.js, .NET Core

Web & Mobile Application Development — HTML5, CSS3, React.js, React Native

Database Management — SQL, PostgreSQL, MongoDB, Firebase, Supabase

DevOps & Cloud Integration — Docker, Kubernetes, Git, Postman, Linux, DBeaver, RabbitMQ, CI/CD, AWS

Machine Learning & Automation — PyTorch, NLP, Computer Vision, OCR, UiPath, Microsoft Power Automate

Software Engineering — Agile Development, API Design & Management, DSA, Microservices, MVC, OOP, Testing

Soft Skills — Problem Solving, Team Collaboration, Adaptability, Time Management, and Eagerness to Grow

KEY PROJECTS

- 05/2026 – Present **AI-Assisted Microservice Incident Monitoring & Root Cause Analysis Platform** 
- Technologies: ASP.NET Core, RabbitMQ, PostgreSQL, Redis, Docker, FastAPI, React, DBeaver, Git, Grafana, Prometheus
- Architected a cloud-native, event-driven microservices platform for intelligent incident monitoring, automated root cause analysis, and resilient inter-service communication across distributed environments.
 - Built asynchronous processing pipelines with RabbitMQ and Redis while integrating AI-powered FastAPI inference services for real-time log analytics, fault detection, caching, and operational observability.
- 09/2025 – 03/2026 **Lightweight Unified Low-Light Enhancement and Classification System** 
- Technologies: Python, FastAPI, PyTorch, OpenCV, Scikit-learn, MobileNetV3
- Developed a unified computer vision pipeline combining low-light enhancement and image classification for real-time inference under constrained environments.
 - Structured a modular processing architecture with adaptive optimization techniques.
- 05/2025 – 06/2025 **PETMEDIX – Pet Healthcare and Management System** 
- Technologies: MERN Stack, ResNet, REST APIs, Jitsi API, Nodemailer
- Architected a scalable full-stack healthcare platform integrating ML-based disease prediction, modular REST APIs, and structured multi-entity data management.
 - Implemented real-time consultation, asynchronous notification handling, and event-driven backend workflows for reliable end-to-end system operations.
- 01/2025 – 03/2025 **Automated Invoice Processing & Bank Record Management System** 
- OCR, SQL, API Integration, RE Framework, Python
- Engineered an intelligent invoice processing system for automated extraction and validation of handwritten and typed PDF data into structured banking records.
 - Designed OCR-driven transformation and validation workflows with relational database integration for high-accuracy enterprise financial processing.

PUBLICATIONS

- 26/03/2026 **Lightweight Unified Network for Low-Light Image Enhancement and Image Classification** 
- 2026 IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE)

CERTIFICATES

Professional Certificate in Machine Learning
Offered by IIT

Professional Certificate in Network Administration
Offered by IIT

Professional Certificate in Cyber Security
Offered by IIT

REFERENCES

Mr. Harindra De Silva, *Senior Software Engineer*, NTB Bank
harindra98@gmail.com, +9476 824 7712

Ms. Hiruni Samarage, *Lecturer*, IIT
hiruni.s@iit.ac.lk, +9476 8209 721